ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jan 2022 P2HR Worked Solutions

Jan 2022 | P2HR

27 questions ▪ question image followed by worked solution

Prepared by

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Student Edition

Worked answers included

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PAST PAPER QUESTION

Q#: 1 Marks: 4

TOPIC: **Probability Toolkit**

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1 A tin contains tea bags with a choice of four different flavours of tea.
The four flavours of tea are Assam or Darjeeling or Nilgiri or Rize.

Sara takes at random a tea bag from the tin.

The table shows each of the probabilities that the flavour of the tea Sara takes is Assam or Darjeeling or Rize.

Flavour of tea	Assam	Darjeeling	Nilgiri	Rize
Probability	0.38	0.24		0.16

(a) Work out the probability that the flavour of the tea Sara takes is Nilgiri.

....

(2)

(b) Work out the probability that the flavour of the tea Sara takes is either Darjeeling or Rize.

....

(2)

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 4

TOPIC: **Probability Toolkit**

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WORKED SOLUTION

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Topic check. Probability Toolkit. The tag is correct.

Solution

The missing probability is

$$1 - 0.38 - 0.24 - 0.16 = 0.22$$

So

$$P(\text{Darjeeling or Rize}) = 0.24 + 0.16 = 0.40$$

FINAL ANSWER

Nilgiri 0.22, Darjeeling or Rize 0.40.

PAST PAPER QUESTION

Q#: 2 Marks: 7

Topic: **Percentages**

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2 Mary saves for a holiday each year.

In 2020 she saved a total of \$720
In 2021, each month she saved \$78

The total amount Mary saved in 2021 was P% more than the total she saved in 2020

(a) Work out the value of P

....
(4)

Roberto is going to go on holiday.

He has two coupons that will save him money on his holiday.

Coupon A	**Coupon B**
18% off the cost of the accommodation	12.5% off the total cost of the accommodation **and** the flights

For Roberto's holiday
the cost of the accommodation is \$1600
the cost of the flights is \$800

Roberto can only use one of the coupons.
He wants to save as much money as he can.

(b) Which of the two coupons, **A** or **B**, should he use?
Show your working clearly.

(3)

(Total for Question 2 is 7 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 7

TOPIC: **Percentages**

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WORKED SOLUTION

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Topic check. Percentages. The tag is correct.

Method

Mary saved in 2021:

$$78 \times 12 = 936$$

Increase from 2020:

$$936 - 720 = 216$$

Percentage increase:

$$\frac{216}{720} \times 100 = 30$$

So $P = 30$.
Coupon A saves

$$18\% \times 1600 = 0.18 \times 1600 = 288$$

Coupon B saves 12.5% of the accommodation and flights together:

$$1600 + 800 = 2400$$

$$12.5\% \times 2400 = 0.125 \times 2400 = 300$$

Coupon B saves more money.

FINAL ANSWER
(a) $P = 30$, (b) Coupon B

PAST PAPER QUESTION Q#: 3 Marks: 5

TOPIC: **Solving Linear Equations**

Jan 2022 P2HR - Jan 2022 | P2HR

3 (a) Solve $4y + 5 > 12$

....
(2)

(b) Solve $6x - 5 = \dfrac{4x - 7}{2}$

Show clear algebraic working.

$x =$ ....
(3)

(Total for Question 3 is 5 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 5

TOPIC: **Solving Linear Equations**

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Topic check. Corrected to solving linear equations. The linear equation is the larger part of the question.

Solution

(a)

$$4y + 5 > 12$$

$$4y > 7$$

$$y > \frac{7}{4}$$

(b)

$$6x - 5 = \frac{4x - 7}{2}$$

Multiply by 2:

$$12x - 10 = 4x - 7$$

$$8x = 3$$

$$x = \frac{3}{8}$$

FINAL ANSWER

$y > \frac{7}{4}$, and $x = \frac{3}{8}$.

PAST PAPER QUESTION

Topic: **Angles in Polygons & Parallel Lines**

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Q#: 4 **Marks: 4**

4 The diagram shows a regular octagon *ABCDEFGH* and a regular pentagon *ABIJK*

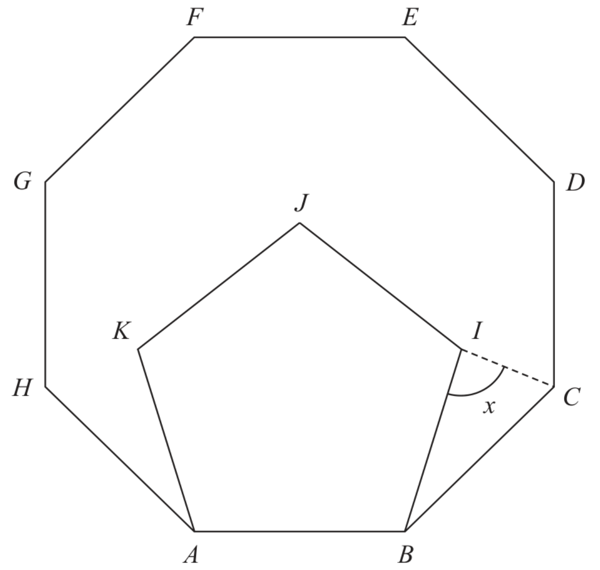

Diagram **NOT** accurately drawn

Work out the size of the angle x

.... °

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

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Topic check. Angles in Polygons & Parallel Lines. The tag is correct.

Solution

The interior angle of a regular octagon is

$$\frac{(8-2)\times 180}{8} = 135^\circ$$

The interior angle of a regular pentagon is

$$\frac{(5-2)\times 180}{5} = 108^\circ$$

At B,

$$\angle IBC = 135^\circ - 108^\circ = 27^\circ$$

Since both polygons share side AB,

$$BI = BC$$

So triangle BIC is isosceles.

$$x = \frac{180^\circ - 27^\circ}{2} = 76.5^\circ$$

FINAL ANSWER

76.5°.

PAST PAPER QUESTION Q#: 5 Marks: 3

TOPIC: **Compound Interest & Depreciation**

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5 Shane invests 7200 dollars for 3 years in a savings account.
He gets 2.5% per year compound interest.

How much money will Shane have in his savings account at the end of 3 years?
Give your answer to the nearest dollar.

.... dollars

(Total for Question 5 is 3 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 3

TOPIC: **Compound Interest & Depreciation**

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Topic check. Compound Interest and Depreciation. The tag is correct.

Method

The multiplier for 2.5% compound interest is

$$1.025$$

After 3 years,

$$7200(1.025)^3 = 7753.6125$$

Correct to the nearest dollar,

$$7753.6125 \approx 7754$$

FINAL ANSWER

7754 dollars

PAST PAPER QUESTION Q#: 6 Marks: 4

TOPIC: **Algebraic Roots & Indices**

Jan 2022 P2HR - Jan 2022 | P2HR

6 (a) Write down the value of x^0

....
(1)

Given that $2^{-3} \times 2^9 = 2^n$

(b) find the value of n

$n =$ ....
(1)

Given that $\dfrac{7^{206} \times 7^m}{7^{214}} = 7^{-3}$

(c) find the value of m

$m =$ ....
(2)

(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 4

TOPIC: **Algebraic Roots & Indices**

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WORKED SOLUTION

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Topic check. Algebraic Roots and Indices. The tag is correct.

Method

$$x^0 = 1$$

For part (b),

$$2^{-3} \times 2^9 = 2^{-3+9} = 2^6$$

So

$$n = 6$$

For part (c),

$$\frac{7^{206} \times 7^m}{7^{214}} = 7^{206+m-214}$$

This equals 7^{-3}, so

$$206 + m - 214 = -3$$

$$m = 5$$

FINAL ANSWER
(a) 1, (b) $n = 6$, (c) $m = 5$.

PAST PAPER QUESTION

Q#: 7 Marks: 5

TOPIC: **Graphing Inequalities**

Jan 2022 P2HR - Jan 2022 | P2HR

7 (a) Write down an equation of the straight line with gradient –3 and which passes through the point with coordinates (0, 5)

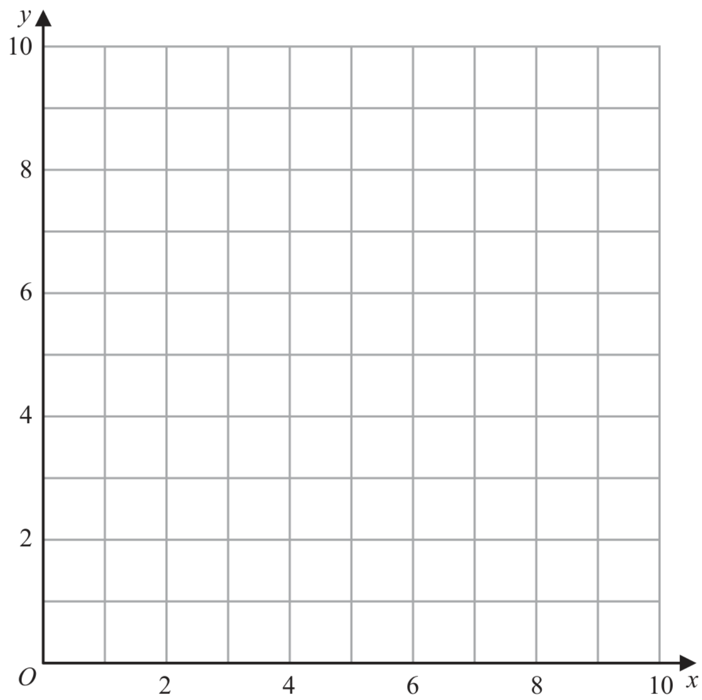
....

(2)

(b) Show, by shading on the grid, the region defined by **all three** of the inequalities

$x \leqslant 6$ $\qquad$ $y \geqslant 2$ $\qquad$ $y \leqslant x + 1$

Label the region **R**



(3)

(Total for Question 7 is 5 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 5

TOPIC: **Graphing Inequalities**

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Topic check. Correct. This is drawing and shading inequalities.

Solution

A straight line with gradient -3 passing through $(0, 5)$ has equation

$$y = -3x + 5$$

For the region, use the inequalities:

$$x < 6$$

$$y > 2$$

$$y \leq x + 1$$

FINAL ANSWER

(a) $y = -3x + 5$. (b) Shade $x < 6$, $y > 2$, $y \leq x + 1$.

PAST PAPER QUESTION Q#: 8 Marks: 3

TOPIC: **Statistics Toolkit**

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8 A scientist is investigating the weight of 50 tigers.

Here is some information about these tigers.

	Type of tiger	
	Siberian	**Bengal**
Number of tigers	22	28
Mean weight of tigers (kg)	260	

The mean weight of all 50 tigers is 218 kg

Work out the mean weight of the Bengal tigers.

.... kg

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 3

TOPIC: **Statistics Toolkit**

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Topic check. Statistics Toolkit. The tag is correct.

Solution

Total mass of all 50 tigers:

$$50 \times 218 = 10900$$

Total mass of the 22 Siberian tigers:

$$22 \times 260 = 5720$$

Total mass of the 28 Bengal tigers:

$$10900 - 5720 = 5180$$

Mean mass of the Bengal tigers:

$$\frac{5180}{28} = 185$$

FINAL ANSWER

185 kg.

PAST PAPER QUESTION Q#: 9 Marks: 5

TOPIC: **Circles, Arcs & Sectors**

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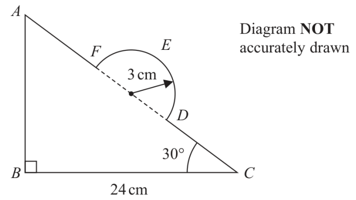
9 In the diagram, ABC is a right-angled triangle and DEF is a semicircular arc.



In triangle ABC

$BC = 24$ cm   angle $ABC = 90°$ angle $BCA = 30°$

The points D and F lie on AC so that DF is the diameter of the semicircular arc DEF
The radius of the semicircular arc is 3 cm.

Work out the length of $AFEDC$
Give your answer correct to 2 significant figures.

.... cm

(Total for Question 9 is 5 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 5

TOPIC: **Circles, Arcs & Sectors**

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Topic check. Corrected to Circles, Arcs & Sectors.

Solution

In triangle ABC,

$$\cos 30^\circ = \frac{24}{AC}$$

$$AC = \frac{24}{\cos 30^\circ} = 27.712\ldots$$

The semicircle has radius 3 cm, so its diameter is 6 cm.

$$AF + DC = AC - DF = 27.712\ldots - 6$$

Arc FED is a semicircle:

$$\text{arc length} = 3\pi$$

Therefore

$$AFEDC = 27.712\ldots - 6 + 3\pi = 31.137\ldots$$

FINAL ANSWER

31 cm.

PAST PAPER QUESTION Q#: 10 Marks: 3

TOPIC: **Powers, Roots & Standard Form**

Jan 2022 P2HR - Jan 2022 | P2HR

10 The table gives information about the population and the total amount of money, in dollars, spent on healthcare for two countries in 2016

Country	**Total population**	**Total spent on healthcare ($)**
Austria	8.7×10^6	4.2×10^{10}
Luxembourg	6.3×10^5	3.7×10^9

Work out how much more was spent **per person** on healthcare in Luxembourg than in Austria.
Give your answer correct to the nearest whole number.

.... dollars

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 3

TOPIC: **Powers, Roots & Standard Form**

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Topic check. Powers, roots and standard form. The tag is correct.

Method

Austria spending per person:

$$\frac{4.2 \times 10^{10}}{8.7 \times 10^{6}} = 4827.586\ldots$$

Luxembourg spending per person:

$$\frac{3.7 \times 10^{9}}{6.3 \times 10^{5}} = 5873.015\ldots$$

Difference:

$$5873.015\ldots - 4827.586\ldots = 1045.429\ldots$$

To the nearest whole number, this is

$$1045$$

FINAL ANSWER

1045 dollars

PAST PAPER QUESTION

Q#: 11 Marks: 4

TOPIC: **Graphs of Functions**

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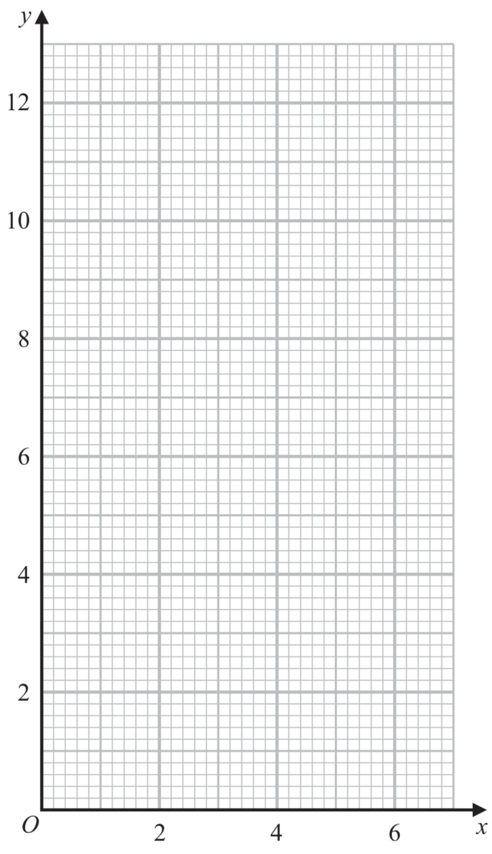
11 (a) Complete the table of values for $y = \frac{6}{x}$

x	0.5	1	2	3	4	5	6
y		6		2			1

(2)

(b) On the grid, draw the graph of $y = \frac{6}{x}$ for $0.5 \leqslant x \leqslant 6$



(2)

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 4

TOPIC: **Graphs of Functions**

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Topic check. Correct. This is drawing a reciprocal graph.

Solution

$$y = \frac{6}{x}$$

The completed table is:

x	0.5	1	2	3	4	5	6
y	12	6	3	2	1.5	1.2	1

Plot the points and join them with a smooth decreasing reciprocal curve.

FINAL ANSWER

Missing values are $12, 3, 1.5, 1.2$.

PAST PAPER QUESTION

Q#: 12 Marks: 3

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2022 P2HR - Jan 2022 | P2HR

12 The diagram shows two vertical phone masts, *AB* and *CD*, on horizontal ground.

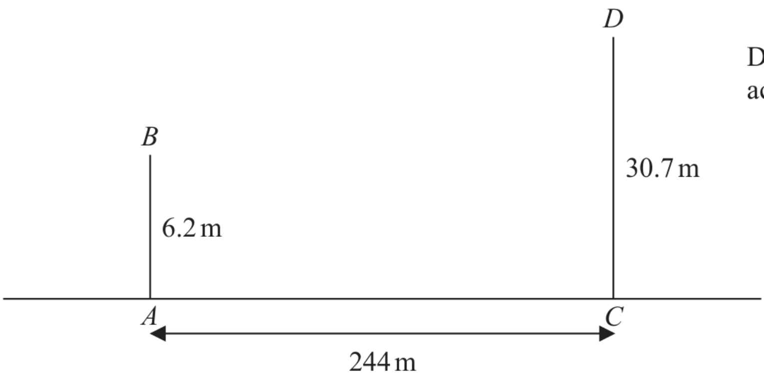

Diagram **NOT** accurately drawn

$AB = 6.2\,\text{m} \qquad AC = 244\,\text{m} \qquad CD = 30.7\,\text{m}$

Work out the size of the angle of depression of *B* from *D*
Give your answer correct to one decimal place.

....°

(Total for Question 12 is 3 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 3

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

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Topic check. Right-Angled Triangles - Pythagoras & Trigonometry. The tag is correct.

Solution

The vertical difference between the tops of the masts is

$$30.7 - 6.2 = 24.5 \text{ m}$$

The horizontal distance is 244 m.

$$\tan\theta = \frac{24.5}{244}$$

$$\theta = 5.733\ldots^\circ$$

FINAL ANSWER

5.7°.

PAST PAPER QUESTION Q#: 13 Marks: 3

TOPIC: **Algebraic Proof**

Jan 2022 P2HR - Jan 2022 | P2HR

13 $a = \sqrt{8} + 4$

$b = \sqrt{8} - 4$

$(a - b)(a + b)$ can be written in the form $y\sqrt{4y}$

Find the value of y

Show your working clearly.

$y =$ ....

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 3

TOPIC: **Algebraic Proof**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION

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Topic check. Algebraic proof. The tag is acceptable because the question asks for exact algebraic working.

Solution

$$a = \sqrt{8} + 4, \qquad b = \sqrt{8} - 4$$

$$a - b = (\sqrt{8} + 4) - (\sqrt{8} - 4) = 8$$

$$a + b = (\sqrt{8} + 4) + (\sqrt{8} - 4) = 2\sqrt{8} = 4\sqrt{2}$$

So

$$(a - b)(a + b) = 8 \times 4\sqrt{2} = 32\sqrt{2}$$

The question says this equals $y\sqrt{4y}$. Try $y = 8$:

$$8\sqrt{32} = 8 \times 4\sqrt{2} = 32\sqrt{2}$$

FINAL ANSWER

$y = 8$.

PAST PAPER QUESTION

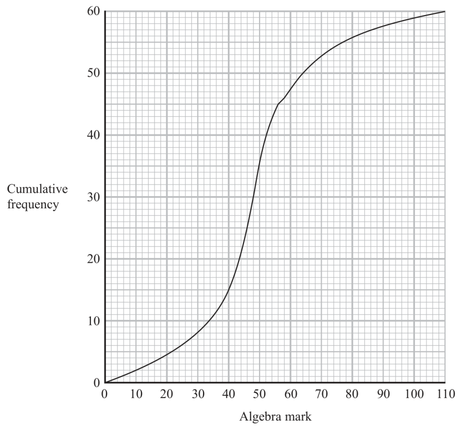
Q#: 14 Marks: 8

TOPIC: **Cumulative Frequency Diagrams**

Jan 2022 P2HR - Jan 2022 | P2HR

14 A group of 60 students each sat an algebra test and a geometry test.
Each test was marked out of 110

The cumulative frequency graph gives information about the marks gained by the 60 students in the algebra test.



(a) Use the graph to find an estimate for the median mark in the algebra test.

....
(1)

(b) Use the graph to find an estimate for the number of students who gained 58 marks or less in the algebra test.

....
(1)

(c) Use the graph to find an estimate for the interquartile range of the marks gained in the algebra test.

....
(2)

The interquartile range of the marks gained in the geometry test is 9

Luis says

"The students' marks are more spread out in the algebra test than in the geometry test."

(d) Is Luis correct?
Give a reason for your answer.

....

....
(1)

To be awarded a grade A in the algebra test, a student had to gain a mark greater than 64

Two students are to be selected at random from the 60 students in the group.

(e) Use the graph to find an estimate for the probability that both of these students were awarded a grade A in the algebra test.

....
(3)

(Total for Question 14 is 8 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 8

TOPIC: **Cumulative Frequency Diagrams**

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Topic check. Cumulative Frequency Diagrams. The tag is correct.

Solution

There are 60 students.
The median is at cumulative frequency 30. From the graph,

$$\text{median} \approx 49$$

At 58 marks, the cumulative frequency is about 45, so about 45 students gained 58 marks or less.
For the interquartile range:

$$Q_1 \text{ is at } 15, \qquad Q_3 \text{ is at } 45$$

From the graph,

$$Q_1 \approx 40, \qquad Q_3 \approx 57$$

$$\text{IQR} \approx 57 - 40 = 17$$

The geometry IQR is 9, so Luis is correct because the algebra IQR is larger.
For grade A, use mark 64. From the graph, about 50 students gained 64 or less, so about

$$60 - 50 = 10$$

students gained more than 64.
Probability both selected students were grade A:

$$\frac{10}{60} \times \frac{9}{59} = 0.0254\ldots$$

FINAL ANSWER

median about 49, about 45 students, IQR about 17, Luis is correct, probability about 0.025.

PAST PAPER QUESTION **Q#: 15** **Marks: 4**

TOPIC: **Rearranging Formulas**

Jan 2022 P2HR - Jan 2022 | P2HR

15 Make t the subject of $n^2 = \dfrac{4d + t^3}{t^3}$

....

(Total for Question 15 is 4 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 4

TOPIC: **Rearranging Formulas**

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Topic check. Rearranging formulas. The tag is correct.

Solution

$$n^2 = \frac{4d + t^3}{t^3}$$

$$n^2 t^3 = 4d + t^3$$

$$t^3(n^2 - 1) = 4d$$

$$t^3 = \frac{4d}{n^2 - 1}$$

FINAL ANSWER

$t = \sqrt[3]{\frac{4d}{n^2-1}}$

PAST PAPER QUESTION

Q#: 16 Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2022 P2HR - Jan 2022 | P2HR

16 The diagram shows quadrilateral *ABCD*

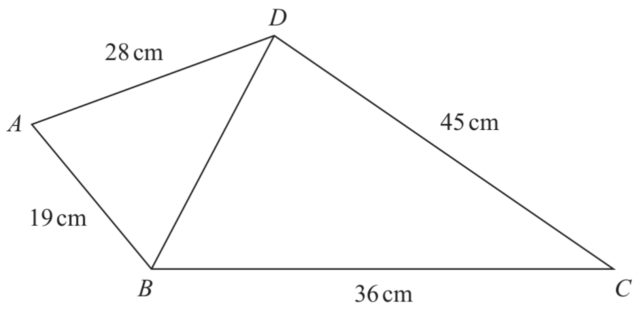

Diagram **NOT** accurately drawn

The angle *BCD* is acute.

Given that the area of triangle $BCD = 405\,\text{cm}^2$

work out the size of angle *ABD*
Give your answer correct to one decimal place.

....°

(Total for Question 16 is 5 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

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Topic check. Sine, Cosine Rule & Area of Triangles. The tag is correct.

Solution

For triangle BCD,

$$405 = \frac{1}{2}(36)(45)\sin BCD$$

$$\sin BCD = 0.5$$

The angle BCD is acute, so

$$BCD = 30^\circ$$

Find BD:

$$BD^2 = 36^2 + 45^2 - 2(36)(45)\cos 30^\circ$$

$$BD = 22.695\ldots$$

Now use cosine rule in triangle ABD:

$$\cos ABD = \frac{19^2 + BD^2 - 28^2}{2(19)(BD)}$$

$$ABD = 83.871\ldots^\circ$$

FINAL ANSWER

83.9°.

PAST PAPER QUESTION

Q#: 17 Marks: 3

TOPIC: **Estimating Gradients**

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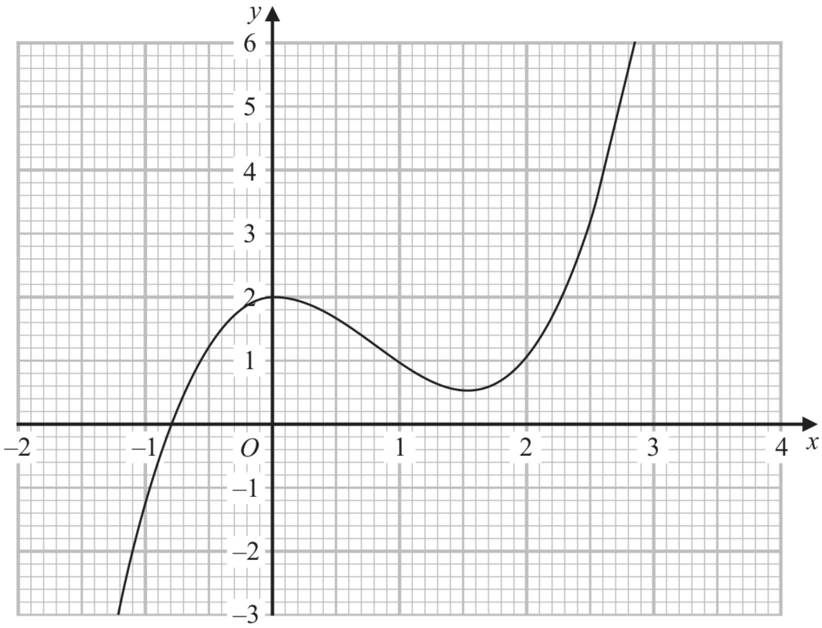
17 Part of the curve with equation $y = f(x)$ is shown on the grid.



Find an estimate for the gradient of the curve at the point where $x = 2$
Show your working clearly.

....

(Total for Question 17 is 3 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 3

TOPIC: **Estimating Gradients**

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Topic check. Corrected to estimating gradients. The question asks for an estimated gradient from a curve.

Solution

Draw a tangent to the curve at the point where

$$x = 2$$

Using two clear points on the tangent, for example approximately $(1.5, 0.6)$ and $(2.3, 2.2)$,

$$\text{gradient} \approx \frac{2.2 - 0.6}{2.3 - 1.5}$$

$$\text{gradient} \approx 2.0$$

FINAL ANSWER

The gradient is approximately 2.

PAST PAPER QUESTION Q#: 18 Marks: 5

TOPIC: **Coordinate Geometry**

Jan 2022 P2HR - Jan 2022 | P2HR

18 The line with equation $2y = x + 1$ intersects the curve with equation $3y^2 + 7y + 16 = x^2 - x$ at the points A and B

Find the coordinates of A and the coordinates of B
Show clear algebraic working.

(.......................... , ...) and (.......................... , ...)

(Total for Question 18 is 5 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 5

TOPIC: **Coordinate Geometry**

Jan 2022 P2HR - Jan 2022 | P2HR

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Topic check. Coordinate geometry. The tag is correct.

Solution

The line is

$$2y = x + 1$$

So

$$x = 2y - 1$$

Substitute into the curve

$$3y^2 + 7y + 16 = x^2 - x$$

$$3y^2 + 7y + 16 = (2y - 1)^2 - (2y - 1)$$

$$3y^2 + 7y + 16 = 4y^2 - 6y + 2$$

$$y^2 - 13y - 14 = 0$$

$$(y - 14)(y + 1) = 0$$

So

$$y = 14 \quad \text{or} \quad y = -1$$

If $y = 14$, then $x = 27$.
If $y = -1$, then $x = -3$.

FINAL ANSWER
$(27, 14)$ and $(-3, -1)$.

PAST PAPER QUESTION

Q#: 19 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Jan 2022 P2HR - Jan 2022 | P2HR

19 *ABCD* is a horizontal rectangular field.

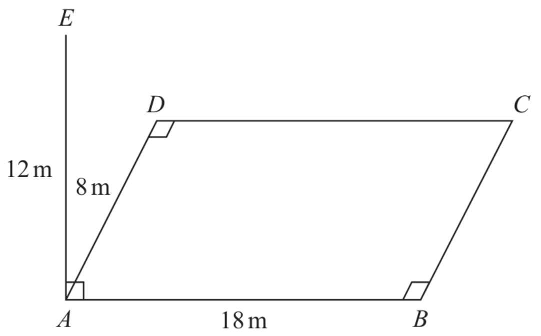

Diagram **NOT** accurately drawn

A vertical pole, *AE*, is placed at the corner *A* of the field.

$AE = 12\,\text{m}$ $\quad AB = 18\,\text{m}$ $\quad AD = 8\,\text{m}$

Calculate the size of the angle between *EC* and the plane *ABCD*
Give your answer correct to one decimal place.

....°

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION Q#: 19 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

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WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. 3D Pythagoras & Trigonometry. The tag is correct.

Solution

The projection of EC onto the field is AC.

$$AC = \sqrt{18^2 + 8^2} = \sqrt{388}$$

The vertical height is $AE = 12$ m, so

$$\tan\theta = \frac{12}{\sqrt{388}}$$

$$\theta = 31.350\ldots^\circ$$

FINAL ANSWER

31.4°.

PAST PAPER QUESTION

Q#: 20 **Marks: 4**

TOPIC: **Direct & Inverse Proportion**

Jan 2022 P2HR - Jan 2022 | P2HR

20 y is inversely proportional to $\sqrt{x}$
x is directly proportional to T^3

Given that $y = 8$ when $T = 25$

find the exact value of T when $y = 27$

$T =$ ....

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION Q#: 20 Marks: 4

TOPIC: **Direct & Inverse Proportion**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Direct and Inverse Proportion. The tag is correct.

Method

y is inversely proportional to $\sqrt{x}$, so

$$y \propto \frac{1}{\sqrt{x}}$$

x is directly proportional to T^3, so $\sqrt{x} \propto T^{3/2}$.
Therefore

$$y \propto \frac{1}{T^{3/2}}$$

So

$$yT^{3/2} = k$$

Use $y = 8$ when $T = 25$:

$$k = 8 \times 25^{3/2} = 8 \times 125 = 1000$$

When $y = 27$,

$$27T^{3/2} = 1000$$

$$T^{3/2} = \frac{1000}{27} = \left(\frac{10}{3}\right)^3$$

Raise both sides to the power $\frac{2}{3}$:

$$T = \left(\frac{10}{3}\right)^2 = \frac{100}{9}$$

FINAL ANSWER
$T = \frac{100}{9}$.

PAST PAPER QUESTION

Q#: 20 Marks: 4

TOPIC: **Direct & Inverse Proportion**

Jan 2022 P2HR - Jan 2022 | P2HR

20 y is inversely proportional to $\sqrt{x}$
x is directly proportional to T^3

Given that $y = 8$ when $T = 25$

find the exact value of T when $y = 27$

$T =$ ....

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Direct & Inverse Proportion**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Direct & Inverse Proportion is correct.

Method

$$y \propto \frac{1}{\sqrt{x}}$$

and

$$x \propto T^3$$

So

$$\sqrt{x} \propto T^{3/2}$$

Therefore

$$y \propto \frac{1}{T^{3/2}}$$

So $yT^{3/2}$ is constant.
When $y = 8$ and $T = 25$,

$$8(25)^{3/2} = 8(125) = 1000$$

When $y = 27$,

$$27T^{3/2} = 1000$$

$$T^{3/2} = \frac{1000}{27} = \left(\frac{10}{3}\right)^3$$

$$T = \left(\frac{10}{3}\right)^2 = \frac{100}{9}$$

FINAL ANSWER

$\frac{100}{9}$

PAST PAPER QUESTION Q#: 21 Marks: 6

TOPIC: **Standard & Compound Units**

Jan 2022 P2HR - Jan 2022 | P2HR

21 The diagram shows a solid made from a cylinder and a hemisphere.
The cylinder and the hemisphere are both made from the same metal.

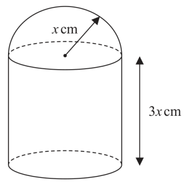


Diagram **NOT** accurately drawn

The plane face of the hemisphere coincides with the upper plane face of the cylinder.

The radius of the cylinder and the radius of the hemisphere are both x cm.
The height of the cylinder is $3x$ cm.

The total surface area of the solid is $81\pi\,\text{cm}^2$
The mass of the solid is 840 grams.

The following table gives the density of each of four metals.

Metal	Density (g/cm^3)
Aluminium	2.7
Nickel	8.9
Gold	19.3
Silver	10.5

The metal used to make the solid is one of the metals in the table.

Determine the metal used to make the solid.
Show your working clearly.

(Total for Question 21 is 6 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 6

TOPIC: **Standard & Compound Units**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The outside surface area is:
curved surface area of cylinder:

$$2\pi x(3x) = 6\pi x^2$$

bottom of cylinder:

$$\pi x^2$$

curved surface area of hemisphere:

$$2\pi x^2$$

So

$$6\pi x^2 + \pi x^2 + 2\pi x^2 = 81\pi$$

$$9\pi x^2 = 81\pi$$

$$x^2 = 9$$

$$x = 3$$

Volume of the cylinder:

$$\pi x^2(3x) = 3\pi x^3$$

Volume of the hemisphere:

$$\frac{2}{3}\pi x^3$$

Total volume:

$$3\pi(3)^3 + \frac{2}{3}\pi(3)^3$$

$$= 81\pi + 18\pi = 99\pi$$

Density:

$$\frac{840}{99\pi} = 2.70\ldots$$

This matches aluminium, whose density is 2.7 g/cm^3.

FINAL ANSWER
Aluminium

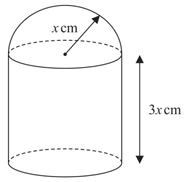
PAST PAPER QUESTION

Q#: 21 **Marks: 6**

TOPIC: **Standard & Compound Units**

Jan 2022 P2HR - Jan 2022 | P2HR

21 The diagram shows a solid made from a cylinder and a hemisphere.
The cylinder and the hemisphere are both made from the same metal.



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Show your working clearly.

(Total for Question 21 is 6 marks)

PAST PAPER SOLUTION

Q#: 21 **Marks: 6**

TOPIC: **Standard & Compound Units**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The outside surface area is:
curved surface area of cylinder:

$$2\pi x(3x) = 6\pi x^2$$

bottom of cylinder:

$$\pi x^2$$

curved surface area of hemisphere:

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So

$$6\pi x^2 + \pi x^2 + 2\pi x^2 = 81\pi$$

$$9\pi x^2 = 81\pi$$

$$x^2 = 9$$

$$x = 3$$

Volume of the cylinder:

$$\pi x^2(3x) = 3\pi x^3$$

Volume of the hemisphere:

$$\frac{2}{3}\pi x^3$$

Total volume:

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Density:

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This matches aluminium, whose density is 2.7 g/cm^3.

FINAL ANSWER
Aluminium

PAST PAPER QUESTION

Q#: 22 **Marks: 5**

TOPIC: **Coordinate Geometry**

Jan 2022 P2HR - Jan 2022 | P2HR

22 ABC is a triangle in which angle $ABC = 90°$

p and q are integers such that

the coordinates of A are $(p, 10)$
the coordinates of B are $(-1, -5)$
the coordinates of C are $(8, q)$

Given that the gradient of AC is $-\frac{6}{7}$

work out the value of p and the value of q

$p =$ ....

$q =$ ....

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Coordinate Geometry**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The gradient of AC is $-\frac{6}{7}$.

$$\frac{q-10}{8-p} = -\frac{6}{7}$$

$$7(q-10) = -6(8-p)$$

$$7q - 70 = -48 + 6p$$

$$6p - 7q = -22$$

Since $\angle ABC = 90^\circ$, lines AB and BC are perpendicular.

$$\text{gradient of } AB = \frac{10-(-5)}{p-(-1)} = \frac{15}{p+1}$$

$$\text{gradient of } BC = \frac{q-(-5)}{8-(-1)} = \frac{q+5}{9}$$

The product of perpendicular gradients is -1:

$$\frac{15}{p+1} \times \frac{q+5}{9} = -1$$

$$15(q+5) = -9(p+1)$$

$$5(q+5) = -3(p+1)$$

$$3p + 5q = -28$$

Now solve

$$6p - 7q = -22$$

$$3p + 5q = -28$$

Double the second equation:

$$6p + 10q = -56$$

Subtract:

$$17q = -34$$

$$q = -2$$

Then

$$3p + 5(-2) = -28$$

$$3p = -18$$

$$p = -6$$

FINAL ANSWER
$p = -6, \; q = -2$

PAST PAPER QUESTION Q#: 22 Marks: 5

TOPIC: **Coordinate Geometry**

Jan 2022 P2HR - Jan 2022 | P2HR

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Given that the gradient of AC is $-\frac{6}{7}$

work out the value of p and the value of q

$p =$ ....

$q =$ ....

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Coordinate Geometry**

Jan 2022 P2HR - Jan 2022 | P2HR

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Dr. Eslam Ahmed

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Method

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$$3p = -18$$

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FINAL ANSWER
$p = -6, \; q = -2$

PAST PAPER QUESTION

Q#: 23 Marks: 4

TOPIC: **Functions**

Jan 2022 P2HR - Jan 2022 | P2HR

23 The functions f and g are such that

$$f(x) = x + 25 \qquad g(x) = x^2 - 12x$$

The function h is such that $h(x) = fg(x)$

The domain of h is $\{x : x \leqslant 6\}$

Express the inverse function h^{-1} in the form $h^{-1}(x) = ...$

$h^{-1}(x) =$ ....

(Total for Question 23 is 4 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 4

TOPIC: **Functions**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$f(x) = x + 25$$

$$g(x) = x^2 - 12x$$

$$h(x) = fg(x) = f(g(x))$$

$$h(x) = x^2 - 12x + 25$$

$$h(x) = (x-6)^2 - 11$$

Let

$$y = (x-6)^2 - 11$$

$$y + 11 = (x-6)^2$$

Since the domain of h is $x < 6$, take the negative square root:

$$x - 6 = -\sqrt{y+11}$$

$$x = 6 - \sqrt{y+11}$$

FINAL ANSWER

$h^{-1}(x) = 6 - \sqrt{x+11}$

PAST PAPER QUESTION

Q#: 23 Marks: 4

TOPIC: **Functions**

Jan 2022 P2HR - Jan 2022 | P2HR

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(Total for Question 23 is 4 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 4

TOPIC: **Functions**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

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FINAL ANSWER
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